MAHATMA GANDHI UNIVERSITY, KOTTAYAM
MGU-UGP (HONOURS) Regular Examination October 2025
Third Semester
Discipline Specific Elective (DSE)

MG3DSEBOTA04 – AGRICULTURE AND FLORICULTURE ALLIED STARTUPS
(2024 ADMISSION ONWARDS)

Answer Key

Duration: 2 Hours **Maximum Marks: 70**

Part A

Short Answer Questions
Answer any 10 questions
Each question carries 2 marks

1. Describe offsets. Give an example.

   Ans: Offsets are small daughter plants that develop from the base of the parent plant. They can be separated and planted to grow independently. Eg. Eichhornia, Pistia

2. Name any two asexual propagation methods.

   Ans: Bulbs, tubers, rhizomes, corms, sucker, offset, runner, cuttings (Any two)

3. What is budding?

   Ans: Budding is an asexual plant propagation method where a bud from one plant (scion) is inserted into the stem of another plant (rootstock).

4. Explain cocopeat?

   Ans: Cocopeat is a natural, fibrous material made from coconut husk. It is used as a soil substitute or soil conditioner because it retains moisture, improves aeration, and is eco-friendly.

5. Mention any two advantages of greenhouses.

   Ans: 1. Provides controlled temperature, humidity, and light for plant growth.

   2. Protects plants from pests, heavy rain, wind, and extreme weather.

6. State one temperature-related precaution in transporting live plants.

Ans: Protect the plants from freezing temperatures (typically below , as this can cause ice crystal damage to cell walls and lead to plant death (freezing injury). Use insulated containers. /Avoid shipping during severe heatwaves or freezing weather. Use heat packs in winter or insulated packaging to mitigate cold/heat shock if necessary.

7. Differentiate between packaging methods for seeds and seedlings.

   Ans:

Feature	Seeds	Seedlings (Live Plants)
Moisture Control	Must be Kept **Dry**. Packaging is often moisture-proof/hermetically sealed (e.g., foil pouches, airtight cans, thick polyethylene) to maintain low moisture content.	Must Retain **Moisture**. Packaging requires moist material around the roots (e.g., damp paper, moss, or the original soil/pot) and is secured to prevent soil spillage.
Structure & Air	Packaging is **compact and rigid/flexible** (e.g., paper envelopes, plastic bags) to protect from mechanical damage. Little to no ventilation needed; often sealed.	Packaging must be **sturdy** (e.g., corrugated boxes) with internal support (stakes, trays) to prevent movement. Ventilation holes are crucial to allow the plant to "breathe" and prevent moisture buildup.

8. State one major disease affecting anthurium cultivation.

   Ans: The most common diseases are root rot (caused by overwatering), anthracnose , and bacterial blight . (any one)

9. Why is dehusking an important initial post-harvest step for mature coconuts intended for copra production?

Ans: Dehusking is an important initial post-harvest step for mature coconuts intended for copra production primarily because it significantly speeds up the drying process and reduces transportation costs.

10. What is the importance of identifying a target audience in marketing?

Ans:

1. Improves Marketing: The marketplace is too differentiated. Target audience helps you understand the demographics of the audience that you should target.
2. Helps In Prioritizing The Resources: When you are clear about your target audience, you can easily prioritize the resources used for marketing.

11. What is Integrated pest management?

Ans: Integrated Pest Management (IPM) is an eco-friendly approach to control pests using a combination of methods like biological control, cultural practices, mechanical methods, and minimal use of chemicals. It aims to reduce pest damage while protecting the environment, humans, and beneficial organisms.

12. List any two advantages of vertical farming.

Ans:

- It promises to increase food supply without using more lands. It would reduce the amount of farmland needed for food.
- This practice ensures food security.
- Extension of growing season/no growing season as food can be grown all year round.
- Pest and diseases infestation is almost completely ruled out as plants are grown in an enclosed environment, leading to products without blemish.
- There is higher yield per unit of floor space as plants are stacked vertically.
- There is control over growing conditions.
- Very active development environment for supporting technologies.
- Weather conditions do not affect plants as you do not have to worry about the amount of rainfall or sunlight or when best to plant your crops.
- Climate control.(**Any two**)

Part B

Short Essay Questions
Answer any 8 questions
Each question carries 5 marks

13. Describe the importance of propagules in floriculture.

Ans: Propagules are plant parts used to produce new plants, such as seeds, cuttings, bulbs, suckers, runners, and tubers. Their importance includes:

Mass multiplication: Helps in producing large numbers of ornamental plants quickly.
True-to-type plants: Vegetative propagules ensure offspring are genetically identical to the parent plant (important for maintaining flower color and shape).
Faster flowering: Propagules like bulbs, rhizomes, and cuttings reach maturity faster compared to seeds.
Seasonal availability: Some ornamental plants do not produce viable seeds; hence propagules are the only means of propagation (e.g., chrysanthemum, rose).
Cost-effective for nurseries: Reduces production time and cost, which benefits commercial floriculture.

14. Describe the factors influencing seed dormancy.

Ans: Seed dormancy is a condition where seeds do not germinate even under favorable conditions. Factors include:

Hard seed coat: Prevents water and oxygen entry (e.g., legumes).
Immature embryo: Embryo is not fully developed at the time of seed dispersal.
Chemical inhibitors: Presence of growth-inhibiting substances like abscisic acid or phenolic compounds in seed coat.
Temperature and light: Some seeds need specific temperatures or light conditions to break dormancy (e.g., lettuce).
Environmental conditions: Dryness or lack of moisture can keep seeds dormant.

15. Describe the effect of hormonal treatment (Natural & Artificial) on plant propagation.

Ans: Plant hormones (phytohormones) play a major role in plant propagation.

Auxins (e.g., IAA, IBA, NAA): Promote root formation in stem and leaf cuttings. Used in rooting powders.
Cytokinins (e.g., kinetin, BAP): Encourage shoot formation and delay leaf aging. Used in tissue culture.
Gibberellins: Break seed dormancy and promote seed germination and stem elongation.
Ethylene: Helps in fruit ripening and flower opening.
Artificial hormones: Synthetic forms like IBA and NAA are widely used in nurseries to improve rooting success and survival rate in plant propagation

16. Write brief notes on the advantages and disadvantages of plant growing structures.

Ans: Plant growing structures (such as greenhouses, polyhouses, shade nets) provide controlled conditions.

Advantages:

- Protection from wind, rain, diseases, and pests.
- Extended growing season and higher yield.
- Controlled temperature, light, and humidity.
- Suitable for high-value crops like orchids, roses, and vegetables.

Disadvantages:

- High initial cost of construction.
- Requires technical knowledge and regular maintenance.
- Risk of disease buildup in closed environments.
- Electricity or energy needed for ventilation and temperature control.

17. Explain the factors that should be taken into consideration for the selection of pots for different plant types?

    Ans:

- Plant size
- Type of root system
- Drainage
- Growth Habit of the Plant
- Material of the Pot
- Weight and Stability

18. Explain how different cultivation techniques affect the yield and quality of Chilly. Provide examples.

    Ans: **1. Land Preparation and Soil Management**

- **Effect:**
  Well-prepared soil ensures proper root growth, aeration, and nutrient availability.
- **Example:**
  - Deep ploughing followed by fine tilth improves root penetration and nutrient uptake.
  - Raised bed cultivation prevents waterlogging and root rot in heavy soils, leading to higher yields.

2. Nursery Management and Seedling Quality

- **Effect:**
  Healthy seedlings establish faster, resist pests better, and ensure uniform growth.
- **Example:**
  - Treating seeds with *Trichoderma viride* or fungicides reduces damping-off, leading to better plant stand and yield.

- Using pro-trays and sterilized cocopeat for seedling production improves transplant vigor.

3. Spacing and Plant Density

- **Effect:**
  Optimal spacing allows better light penetration, air circulation, and nutrient distribution.

- **Examples:**

  - **Closer spacing** (e.g., 45 × 30 cm) increases yield per unit area but may reduce fruit size and quality.
  - **Wider spacing** (e.g., 60 × 45 cm) improves fruit size, color, and reduces disease incidence but may lower total yield.

4. Nutrient Management

- **Effect:**
  Balanced fertilization influences vegetative growth, fruit set, color development, and pungency.
- **Examples:**

  - Application of **Nitrogen (N)** enhances vegetative growth, but excess N delays flowering and reduces capsaicin content.
  - **Phosphorus (P)** promotes flowering and fruiting.
  - **Potassium (K)** improves fruit color, thickness, and shelf life.
  - Use of **organic manures** (e.g., farmyard manure or vermicompost) enhances soil microbial activity and quality of fruits.

5. Irrigation Management

- **Effect:**
  Proper irrigation maintains soil moisture, affecting fruit size and pungency.

- **Examples:**

  - **Drip irrigation** ensures efficient water use, uniform fruit development, and higher yields compared to flood irrigation.

6. Mulching

Effect:
Mulching conserves soil moisture, suppresses weeds, and maintains soil temperature.

(Any 5)

19. Write a short note on the significance of foliage ornamental plants with examples like Cypress, Podocarpus, and Thuja.

    Ans: Foliage ornamental plants are grown mainly for their attractive leaves rather than flowers. Their varied shapes, colors, and textures add beauty, freshness, and greenery to indoor and outdoor spaces throughout the year. They improve air quality, create a cooling effect, and enhance the aesthetic appeal of gardens, parks, and landscapes. Many species are also used for hedges, borders, and bonsai due to their dense and evergreen foliage.

Examples:

- **Cypress (Cupressus sp.)** – Valued for its conical shape and evergreen needle-like leaves; commonly used in avenue planting and formal gardens.

- **Podocarpus sp.** – A slow-growing evergreen shrub or tree used for hedges and topiary work.

- **Thuja sp. (Arborvitae)** – Popular for its aromatic foliage and compact growth; ideal for ornamental hedges and screens.

20. Describe the cultivation practices and post-harvesting uses of Jackfruit.

    Ans:

    Cultivation Practices

    Climate and Soil:
    Jackfruit grows well in warm, humid tropical climates with annual rainfall of 150–200 cm. It prefers deep, well-drained loamy soil rich in organic matter and slightly acidic pH (6.0–6.5).

    Propagation:
    It is propagated by **seeds**, but for true-to-type plants, **vegetative methods** such as inarching, budding, or grafting are preferred.

    Planting:
    Pits of about 1 m^3 are dug and filled with a mixture of topsoil and organic manure. Recommended spacing is **8–10 m apart**. Planting is usually done at the onset of monsoon.

    Manuring and Fertilization:
    Application of **FYM (20–25 kg/tree/year)** along with NPK fertilizers promotes healthy growth and early fruiting.

Irrigation:
Regular watering is essential during the dry season, especially for young plants. Mature trees tolerate short dry periods.

Pruning and Training:
Removal of dead and overcrowded branches helps in better sunlight penetration and fruit quality.

Pest and Disease Management:
Common pests include shoot borers and fruit flies; **bagging of fruits** and use of neem-based sprays help reduce damage.

Harvesting:
Fruits are harvested **3–8 months after flowering** when they emit a dull sound on tapping and spines become well developed.

Post-Harvest Uses

Nutritional Use: The ripe bulbs are rich in carbohydrates, vitamins A and C, and are eaten fresh or used in jams, jellies, and desserts.

Vegetable Use: The immature green fruit is cooked as a popular vegetable or used in pickles and curries.

Processed Products: Chips, dehydrated flakes, and jackfruit flour are now commercially produced.

Seed Utilization: Seeds are edible when roasted or boiled and are a good source of starch and protein.

Wood Use: The timber is durable, termite-resistant, and used in furniture and musical instruments.

21. Describe fertigation.

Ans: Fertigation is a highly efficient agricultural technique that combines fertilization and irrigation by injecting soluble fertilizers directly into the water stream of a pressurized irrigation system, most commonly drip irrigation. This method ensures that water and precise doses of nutrients are delivered simultaneously and directly to the active root zone of the crops. The primary advantages of fertigation include a significant increase in nutrient-use efficiency, substantial water savings, and the ability to apply nutrients exactly when the plant needs them for optimal growth, resulting in higher yields and reduced environmental impact compared to traditional broadcast methods.

22. Write a short essay on effect of soil factors on irrigation.

    Ans: Soil acts as the water reservoir for the plants. Its properties govern how often and how much water must be applied.

- Soil Texture (Sand, Silt, Clay):
    - Sandy Soils: Have large pores, a high infiltration rate (water soaks in quickly), and a low water-holding capacity (it drains out quickly). They require frequent, small irrigation applications.
    - Clay Soils: Have very fine pores, a low infiltration rate (water soaks in slowly), and a high water-holding capacity (it stores water for longer). They can handle less frequent, larger applications, but application rates must be slow to prevent runoff.
- Soil Depth: Deeper soils can store more water, meaning they can sustain a crop for longer periods between irrigations compared to shallow soils.
- Topography (Slope): Fields with steep or uneven slopes are prone to runoff and erosion under surface irrigation (like flood). Pressurized systems (sprinkler or drip) are often preferred as they allow for more controlled application on sloped terrain.

Part C

Essay Questions
Answer any 1 question
Each question carries 10 marks

23. Write an essay on formulating potting mixture for vegetables, orchids, anthuriums & indoor plants.

    Ans: A **potting mixture** is a blend of soil and other materials designed to provide plants with optimum physical, chemical, and biological conditions for growth in containers. A good potting mix should be **well-drained, porous, rich in nutrients, and free from pests and pathogens**. The composition of the mixture varies depending on the type of plant grown—whether it is a vegetable, orchid, anthurium, or indoor ornamental plant.

    1. Characteristics of an Ideal Potting Mixture

An ideal potting mix should:

- Ensure **good aeration** for root respiration.
- Maintain adequate **moisture-holding capacity** without waterlogging.
- Supply **essential nutrients** in balanced proportion.
- Have **neutral to slightly acidic pH (6.0–6.8)**.
- Be **light in weight** and **free from weeds, insects, and pathogens**.

Common ingredients used include **garden soil, sand, compost, farmyard manure (FYM), leaf mold, coir pith, vermicompost, perlite, and charcoal.**

2. Potting Mixture for Vegetables

Vegetables grown in pots or grow-bags require a nutrient-rich, well-drained medium.

Recommended mixture:

- Red soil : Sand : FYM or compost = **1 : 1 : 1 (by volume)**

For better aeration, **cocopeat (1 part)** or **perlite (10%)** may be added.
Organic amendments such as neem cake or bone meal improve fertility and pest resistance.
pH: Around 6.0–6.5 is ideal for most vegetables.

Example:
Tomato, brinjal, and chilli thrive well in this balanced mixture, while leafy vegetables like spinach prefer slightly more compost for moisture retention.

Potting Mixture for Orchids (Epiphytes)

Most cultivated orchids, being epiphytes, naturally grow on trees and do not tolerate standing water or fine soil.

Key Needs: Extremely high porosity, rapid drainage, and structural support.

- **Ideal Components:**
    - **Base/Support (90-100%):** Medium to coarse-grade fir bark or pine bark chunks.
    - **Aeration/Retention:** Charcoal (provides aeration and absorbs impurities) and sphagnum moss (used sparingly for moisture retention, especially for smaller-rooted species).

Fine particles like soil or peat must be avoided as they suffocate the roots.

Potting Mixture for Anthuriums

Anthuriums are **semi-epiphytic tropical plants** requiring a **light, porous, and organic-rich** substrate.

Recommended mixture:

- Coconut husk pieces : Charcoal : Brick pieces : Cocopeat : Perlite = **2 : 1 : 1 : 1 : 1**

This mixture ensures proper drainage and humidity around the roots. Anthuriums thrive best in **slightly acidic pH (5.5–6.5)**. Addition of **slow-release NPK fertilizers** or **organic liquid manures** improves flowering and leaf color.

Avoid compact soil, as it leads to root rot and poor aeration.

Potting Mixture for Indoor Plants

Indoor plants grow in limited light and air circulation, so the medium should be **lightweight, moisture-retentive, and well-aerated**.

General mixture:

- Garden soil : Compost : Sand : Cocopeat = **1 : 1 : 1 : 1**

Adding **activated charcoal** helps absorb toxins and odour. **Perlite or vermiculite** may be included to enhance aeration. Periodic top-dressing with organic manure or liquid fertilizer keeps the plants healthy.

Example: Money plant, Syngonium, Dieffenbachia, and Areca palm perform well in this mixture.

24. Write an essay on advantages and disadvantages of intercropping.

    Ans: Intercropping is a method of growing more than one crop in the same piece of land during the same crop season. This is an environmentally friendly pest control, crop protection, and yield enhancement approach.

    Advantages of intercropping

    • Better utilization of growth resources like nutrients, light, and moisture.

    • Mitigate losses where the failure of any one of the component crops occurs.

    • A farmer can get extra yield and, thereby, extra income from the unit area.

    • Better utilization of space is available with the time dimensions.

    • We can achieve better weed control by growing intercrops.

    • Minimizes the chances of soil crust formation.

    • Mitigate losses where the failure of any one of the component crops occurs.

    • Intercrops will also enhance soil fertility.

    • Controls soil erosion.

    • To some extent, the pests and diseases of the main crop are controlled.

    • Ecological stability.

• Cultivation practices for the major crop will also supplement companion crop requirements.

• It provides shading and physical support to some crops.

• It provides daily needs for the farmers.

Disadvantages of intercropping

• Sometimes, intercrops work as alternate hosts for various pests and diseases.

• Differential maturity and sometimes harvesting may become a problem.

• It becomes difficult to control various pests and diseases.

• Intercropping is a labor-intensive practice.

• Allelopathic effect. (Plants produce biochemicals that influence the growth and reproduction of others.)

• Possibility of problems in carrying out intercultural operations.

• Difficulty in mechanization for processes like crop cutting, spraying, and irrigation.

• There might be competition among the component crops.